

UEC619: ELECTROMAGNETIC FIELD THEORY AND TRANSMISSION LINES

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Course Objective: To enhance student's comprehensive capabilities in electromagnetic field theory by studying the behaviour of static and time varying electric and magnetic fields in a medium and transmission line.

Vector Analysis: Coordinate system: Cartesian, Cylindrical and spherical coordinate systems, Vector algebra: dot product, cross product, gradient, divergence and curl.

Electrostatic fields: Introduction to coulomb's law, Gaussian law and its applications in determination of field of spherical and cylindrical geometries, Laplace's and Poisson's equation in various coordinate systems. Boundary conditions at electric interfaces, Method of images and its applications.

Magnetostatics: Introduction to ampere's law, Magnetic vector potential, Magnetic forces, Boundary conditions at magnetic interfaces.

Time Varying Fields and Maxwell's Equations: Maxwell's equation in integral and differential form: For static and time varying fields, For free space, For good conductors, For harmonically varying fields, Continuity of charge, Concept of displacement current.

Electromagnetic wave propagation: Waves in General, Wave Propagation in Lossy Dielectrics, Plane Waves in Free Space, Plane Waves in Good Conductors, Wave Polarization, Poynting theorem and power flow: Energy stored and radiated power, Reflection of a Plane Wave at Normal Incidence/Oblique Incidence

Transmission Lines and Matching Networks: Introduction, Transmission Line Parameters, Transmission Line Equations, Input Impedance, Characteristic impedance, Standing Wave Ratio, and Power, The Smith Chart, transmission line as circuit and matching elements (quarter wave, single stub and double stub)

Guided waves: Waves between parallel plates, transverse electric waves and magnetic waves, characteristics of TE and TM waves, TEM waves, velocity of propagation, Characteristic impedance at radio frequencies, Propagation constant, Attenuation constant and phase constant

Familiarization with standards: IEEE 148-1959, IEEE 1128-1998 and IEEE 1302-1998.

Course Learning Objectives (CLO)

The students will be able to:

1. Apply the concepts of co-ordinate systems and vector calculus.
2. Analyze the concepts of static electric and magnetic fields.
3. Analyze Maxwell's equations in various forms and apply those to engineering problems, while examining wave propagation phenomena across different media and interfaces.
4. Analyze the concepts of basic transmission media.

Text Books

1. Hayt, W.H., Engineering Electromagnetics, Tata McGraw., 8th ed.
2. Kraus, J.D., Electromagnetics, McGraw., 8th ed.

Reference Books

1. Sadiku, M.N.O, Elements of Electromagnetics, Oxford University Press (2009) 4th ed.

UECxxx: MIXED SIGNAL VLSI DESIGN

L	T	P
3	0	0

Course Objective: The course aims to teach advance design techniques for switched capacitor circuits, comparators, data converters (ADCs/DACs) and phase locked loop (PLL). The objective of the course is to design the product level building blocks for VLSI applications.

Syllabus

Sample and hold and trans-linear circuits: performance of sample-and-hold circuits – testing sample and holds, MOS sample-and-hold basics, examples of CMOS sample-and-hold circuits, trans-linear gain Cell.

Switched Capacitor circuits: basic building blocks – op-amps, capacitors, switches, non-overlapping clocks, basic operation and analysis of switched capacitor circuits, resistor equivalence of a switched capacitor, parasitic-sensitive integrator, parasitic-insensitive integrators, signal-flow-graph analysis, noise in switched-capacitor circuits, first-order filters – switch sharing, fully differential filters, biquad filters.

Comparators: comparator specifications – input offset and noise, hysteresis, op-amp as a comparator–input-offset voltage errors, charge-injection errors, latched comparators, latch-mode time constant, latch offset, examples of CMOS comparators.

Data converters: Fundamentals of data converters, Nyquist rate A/D converters: Flash, interpolating, folding flash, SAR, and pipelined architectures, Nyquist rate D/A converters: voltage, current and charge mode converters, hybrid, and segmented converters, Oversampled A/D and D/A converters.

Oscillators and PLL: Ring oscillators, VCO, LC oscillators, basic phase-locked loop architecture, phase detector, loop filter, the PLL in lock, linearized small-signal analysis, jitter and phase noise.

Familiarization with standards: IEEE 1076.1 (VHDL-AMS) for hardware description, IEEE 1500 for embedded core testing, and IEEE 1687 for instrument connectivity.

Course learning outcome (CLO): The student will be able to:

1. Understand the principle of sample-and-hold and trans-linear circuit.
2. Analyze the operation and non-ideal effects of comparators and switched-capacitor based analog circuits.
3. Design different architectures of ADCs and DACs.
4. Understand different architectures of oscillators and phase locked loops.

Text Books

1. Behzad Razavi, Design of Analog CMOS Integrated Circuits, McGraw Hill (2016)
2. Baker, R.J., Li, H.W. and Boyce, D.E., CMOS: Circuit Design, Layout and Simulation, IEEE Press (2007) 2nd ed.
3. Behzad Razavi, Design of CMOS Phase-Locked Loops: From Circuit Level to Architecture Level, Cambridge University Press, 2020.

Reference Books

1. Plassche, Rudy J. Van De, Integrated A-D and D-A Converters, Springer (2007), 2nd ed.
2. Tony Chan Carusone, David A. Johns, Kenneth W. Martin, Analog Integrated Circuit Design, Wiley (2012).
3. Behzad Razavi, Analysis and Design of Data Converters, Cambridge University Press, 2025.